

Shore Defender: A VR Edutainment Game Aiming to Raise Public Awareness on Ocean Conservation and Plastic Recycling

Md Azizul Hakim*

University of Wisconsin-Milwaukee

Md Nahid Hasan†

University of Wisconsin-Milwaukee

Randall Sanders‡

University of Wisconsin-Milwaukee

Jerald Thomas§

University of Wisconsin-Milwaukee

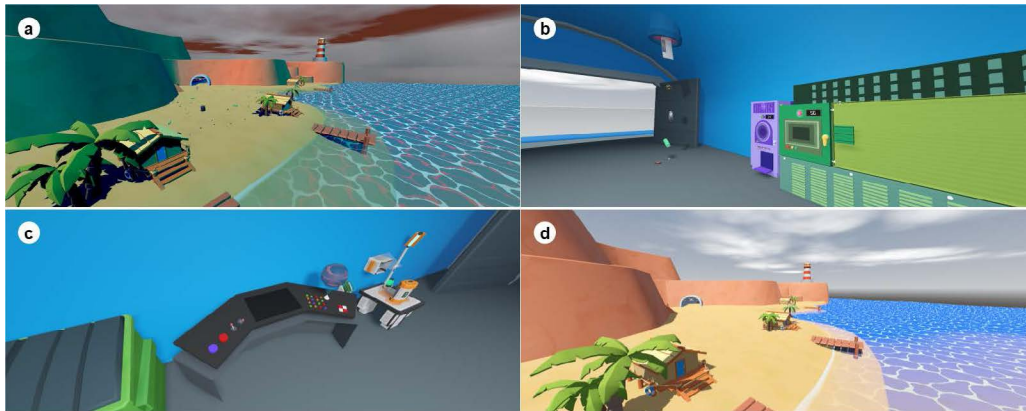


Figure 1: A third-person point-of-view: (a) starting beach; (b) machines in the disposing facility; (c) control panel in the disposing facility; (d) restored beach view upon successful completion of the game.

ABSTRACT

Plastic pollution in marine ecosystems, particularly from beach litter, is a significant threat to ocean conservation and sustainability. This paper presents Shore Defender, a collaborative virtual reality (VR) experience designed to increase public awareness about beach trashing, plastic pollution, and recycling. The game features two asymmetrical roles: one user collects and sorts trash on a virtual beach, while the other processes it in a disposing facility. To enhance the immersive experience, Shore Defender introduces an innovative interaction technique, XR ProMotion, which modifies the traditional scaled-world grab method for extended reaching and engaging gameplay. Through collaborative gameplay, users witness the environmental impact of proper waste management, with significant visual improvements in the virtual environment as they complete the game. By promoting teamwork and environmental stewardship in an engaging virtual setting, Shore Defender aims to inspire real-world action toward ocean conservation.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual Reality; Human-centered computing—Human computer interaction (HCI)—Interaction techniques.

1 INTRODUCTION

Ocean conservation is a critical subject within the broader domain of environmental sustainability. It encompasses the protection of

marine ecosystems, sustainable resource management, and mitigation of anthropogenic impacts. Numerous marine species are facing endangerment due to severe pollution, climate change, and global warming, leading to disruptions in ecological balance and biodiversity. A significant challenge in this effort is plastic pollution. It has been reported that 80% of marine wastes originates from beach littering [2]. Beach litter primarily consists of single-use plastics, such as bottles, straws, and food wrappers. Improperly disposed of single-use plastics degrade into microplastics under the influence of sunlight and wave action, ultimately contributing to the broader issue of marine plastic pollution. This disruption not only affects marine life but also poses significant threats to human livelihoods. Altogether, the problem necessitates increased public awareness and education to mitigate further environmental degradation.

Empirical evidence suggests that enhancing public awareness can lead to meaningful behavioral changes regarding environmental protection. This paper introduces Shore Defender, a multi-user virtual reality (VR) application designed to raise awareness about the detrimental effects of beach littering and plastic pollution. As part of Shore Defender, we propose a novel interaction technique inspired from the traditional scaled-world grab method [4]. This design of interaction enables extended reaching in object grasping and makes the experience more exciting by adding a level of difficulty and required skill. The multi-user experience is provided by designing asymmetric tasks which requires verbal communication between the users. By engaging users in an interactive and immersive virtual environment, Shore Defender aims to foster a deeper understanding of the consequences of pollution and promote responsible waste management practices.

2 STORY AND GAMEPLAY

Shore Defender is meant to be played by two users simultaneously performing asymmetric tasks. As the pair of users progress in their task of cleaning the beach and properly recycling the gathered trash,

*e-mail: hakimm@uwm.edu

†e-mail: hasan26@uwm.edu

‡e-mail: sande333@uwm.edu

§e-mail: jeraldlt@uwm.edu

the environmental becomes visibly cleaner step-by-step. Firstly, the sky is removed of smog and pollution, becoming sunny and pleasantly blue. Secondly, the water loses its dingy grey coloration as more pollution is removed from the environment. Thirdly, the beach gets cleared of debris and other litter. Lastly, the mountains become visibly cleaner. The game is finished once all four stages are complete indicating that the entire environment has been cleaned up.

Beach User At the beginning of the game, users are shown with a polluted, gloomy, and dirty beach. There are 4 types of trash and waste scattered around the beach. They are: organic/food, recyclable, solid, and hazardous wastes. Recyclable wastes are of 4 types, paper, plastic, metal, and glass. The user can freely move along the shore using their left-controller thumbstick that supports a standard direction-based teleportation locomotion technique [1]. The user can also control their virtual rotation in discrete amounts with their right-controller thumbstick. The task of the user is to roam around the beach and collect the trash, which is initially accomplished physically reaching for the trash and grabbing it with the grip button of either controller. Additionally, after picking up and disposing of a certain amount of trash, the user can be equipped with a modified scale-grab selection/manipulation interaction technique we call XR ProMotion, which enables a faster and more entertaining method for collecting the trash. To dispose of a piece of trash the user has to place it in an appropriate bin.

Disposing Facility User The second user in this collaborative game is placed inside a disposing facility. The trash that the first user collected from the beach and placed inside the bins come straight to the user at the disposing facility to be either directly disposed or recycled. To successfully dispose incoming trash, the user needs to place it inside the appropriate machine. After successfully disposing and recycling a number of trash pieces, one button gets activated on the control panel near the machines. By pressing the button, the modified scale-grab technique gets activated for the beach user, thereby making the task simpler for the beach user.

3 INTERACTION TECHNIQUES

To select an object that is out of user's arm reach, we propose an enhanced scaled-world grab interaction technique, referred to as "eXtended Reaching with PROjectile Motion" or "XR ProMotion", which has modified the traditional scaled-world grab manipulation technique to make the experience more challenging and exciting [4]. In the traditional scaled-world grab method, the user first selects an object using a designated selection technique. Once the object is selected, the interface transitions to manipulation mode after instantly teleporting the object to the user's hand, allowing the user to move and rotate the virtual object in 3D space [1].

In our proposed XR ProMotion interaction technique, a raycast starts emitting by holding trigger from either controller and enables selection of selectable objects, including any of the pieces of trash in the environment (Fig. 2). This form of selection is inspired from the traditional scaled-world grab technique. However, our modified version makes the selection subtask easier than the traditional technique. When the laser is over an object, it starts flying towards the user using standard parabolic projectile physics. Nevertheless, in our proposed method, the object does not automatically teleport directly to the user's hand. The user needs to catch the object in the air with regular grasping technique, which is holding the item with grab button with any of their controllers. The XR ProMotion interaction technique can be applied in various scenarios based on the requirements. In our game, it enables extended reaching and makes the experience more exciting.

4 COLLABORATIVE GAMEPLAY

We designed a collaborative VR gameplay that supports asymmetric tasks between two users. The task of the first user is to collect

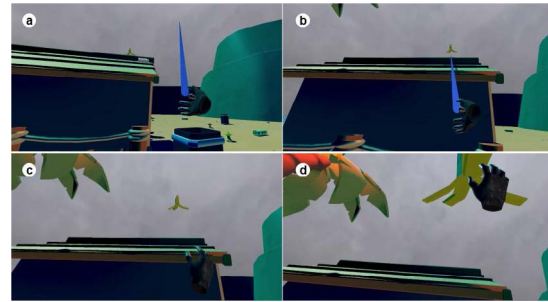


Figure 2: Proposed XR ProMotion interaction technique: (a) the user is unable to reach the desired object (a banana peel in this example); (b) the user selects the object using raycasting; (c) the object flies towards the user using parabolic projectile physics; (d) the user catches the object.

trash from the beach and drop them in the appropriate trash bins that send it from the beach to the factory. On the other hand, the second user's task is to properly dispose of the trash that the first user has sent to him using the bins in real-time. When the first user drops a trash item using any of the bins, it directly teleports to the user at the factory.

The process of disposing the trash depends upon the type of the trash. Non-recyclable wastes need to be either composted or land-filled whereas recyclable wastes need to be recycled by following distinctive processing [3]. To progress the game and finish it in a lower time interval, in-person verbal communication between two users is necessary. It is because of the design of our story which is stated in Section 2.

5 CONCLUSION AND FUTURE WORK

Currently, the task of the user at the disposing facility is limited to simulating abstract way of plastic recycling. We are interested to make a proper simulation of plastic recycling for more gamification of the experience. We also believe that introducing additional tasks, constraints, and challenges will enhance the gameplay and intend to do so in the future.

This project aims to demonstrate the potential of immersive virtual reality experiences in promoting environmental awareness. By engaging users in collaborative and interactive gameplay, it effectively illustrates the impact of pollution and the importance of recycling. Additionally, we proposed an innovative interaction technique which supports extended reaching and grasping in this project.

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